CompTIA Network+ (N10-009)

Chapter 4: Explain Common Networking Ports, Protocols, Services & Traffic Types

A comprehensive study guide to Protocols, Ports, Services and Traffic Types

By Hassen Hannachi | N10-009 Study Series | March 2026

Introduction

Imagine a city full of apartment buildings. Every building has a street address — that's your IP address. But inside each building are hundreds of individual apartments, each with its own number. Those apartment numbers are ports, and the rules the residents follow to communicate are protocols.

Chapter 4 dives into exactly this: how data finds its destination, what rules govern the conversation, and how traffic is classified across modern networks. This is foundational knowledge — mastering it makes the rest of networking click into place.

WHY THIS MATTERS

Ports, protocols, and traffic types are tested directly on the Network+ exam and underpin firewall rules, network troubleshooting, and security analysis in real-world jobs.

Part 1 — Protocols and Ports

Ports — The Doors of Communication

A port is a logical endpoint for communication on a device. When data arrives at your computer via its IP address, the operating system uses the port number to determine which application should handle it.

Port numbers range from 0 to 65,535 and are divided into three official categories defined by IANA (Internet Assigned Numbers Authority):

Port	Protocol	Transport	Description
0 – 1,023	Well-Known Ports	System	Assigned by IANA; require root/admin privileges to bind
1,024 – 49,151	Registered Ports	Vendor	Used by software vendors and registered applications
49,152 – 65,535	Dynamic / Ephemeral	OS	Assigned temporarily by the OS to client-side connections

When your browser opens a webpage, it contacts the server on port 80 (HTTP) or 443 (HTTPS). Your OS simultaneously assigns your browser a random ephemeral port (e.g., 51,423) so the server knows where to send the response back.

MENTAL MODEL

Think of a port like a TV channel. The IP address is the cable box (the device), and the port is the channel number that routes incoming data to the right application (show).

Common Protocols & Port Numbers

A protocol is a standardized set of rules defining how data is formatted, transmitted, and received. The following are the protocols you must know for the exam, grouped by function.

File Transfer & Remote Access

Port	Protocol	Transport	Description
20	FTP (Data)	TCP	File Transfer Protocol — active data channel
21	FTP (Control)	TCP	FTP — command and control channel
22	SSH / SFTP	TCP	Secure Shell — encrypted remote access and file transfer
23	Telnet	TCP	Unencrypted remote access (legacy — avoid in production)
69	TFTP	UDP	Trivial FTP — no authentication, used for booting and configs

Email Protocols

Port	Protocol	Transport	Description
25	SMTP	TCP	Sends email server-to-server (Simple Mail Transfer Protocol)
110	POP3	TCP	Downloads email to client, deletes from server
143	IMAP	TCP	Syncs email; messages stay on server across devices
587	SMTP (Submission)	TCP	Authenticated client-to-server email submission
993	IMAPS	TCP	IMAP over TLS/SSL — encrypted
995	POP3S	TCP	POP3 over TLS/SSL — encrypted

Web & Name Resolution

Port	Protocol	Transport	Description
53	DNS	TCP/UDP	Domain Name System — resolves hostnames to IP addresses
80	HTTP	TCP	HyperText Transfer Protocol — unencrypted web traffic
443	HTTPS	TCP	HTTP over TLS — encrypted web traffic (standard today)

Network Management & Infrastructure

Port	Protocol	Transport	Description
67/68	DHCP	UDP	Dynamic Host Config Protocol — server:67, client:68
123	NTP	UDP	Network Time Protocol — clock synchronization across devices
161/162	SNMP	UDP	Simple Network Mgmt — polling:161, traps:162
389	LDAP	TCP	Lightweight Directory Access Protocol
636	LDAPS	TCP	LDAP over TLS/SSL — encrypted directory access
514	Syslog	UDP	System logging forwarded to a centralized log server
3389	RDP	TCP/UDP	Remote Desktop Protocol — Windows remote GUI session

EXAM TIP

DNS normally uses UDP port 53 for standard queries (fast, lightweight). It switches to TCP port 53 for zone transfers or when a UDP response exceeds 512 bytes. Expect a scenario question about this distinction on the exam.

Part 2 — Internet Protocol Types

At Layer 3 (Network Layer) of the OSI model, several protocols handle how packets are addressed, routed, and delivered. The four IP-level protocols in this section each serve a distinct purpose — and each is tested on the exam.

ICMP — Internet Control Message Protocol

ICMP is not used to transfer user data. Instead, it carries error messages and diagnostic information between network devices. It operates at Layer 3 and is embedded directly in IP packets (protocol number 1).

Common ICMP Message Types

Port	Protocol	Transport	Description
Type 0	Echo Reply	—	Response to a ping request (inbound reply)
Type 3	Destination Unreachable	—	Packet could not reach its destination
Type 5	Redirect	—	Router instructs host to use a more optimal route
Type 8	Echo Request	—	The 'ping' itself — tests host reachability
Type 11	Time Exceeded	—	TTL expired — used by traceroute to map hops

Tools That Use ICMP

- ping — sends ICMP Echo Request (Type 8), awaits Echo Reply (Type 0)
- traceroute / tracert — uses incrementing TTL and ICMP Time Exceeded (Type 11) to map network hops
- pathping — Windows tool combining ping and traceroute with latency statistics

SECURITY NOTE

ICMP is often blocked by firewalls to prevent reconnaissance (e.g., ping sweeps). A lack of ping response does not always mean a host is down — ICMP packets may simply be filtered by a security policy.

TCP — Transmission Control Protocol

TCP is a connection-oriented protocol at Layer 4 (Transport). It guarantees that data arrives completely, in order, and without errors — at the cost of some overhead. TCP is identified by IP protocol number 6.

The Three-Way Handshake

Before any data is sent, TCP establishes a connection using a three-step process:

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Step 1 — SYN
Client → Server:  "I want to connect (seq=100)"

Step 2 — SYN-ACK
Server → Client:  "Acknowledged, I'm ready (seq=200, ack=101)"

Step 3 — ACK
Client → Server:  "Confirmed — connection open (ack=201)"
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Key TCP Features

- Sequencing — packets are numbered; receiver reassembles in correct order
- Acknowledgements — receiver confirms receipt; sender retransmits on timeout
- Flow Control — sliding window mechanism prevents overwhelming the receiver
- Error Checking — checksum field detects corrupted packets in transit
- Connection Termination — four-way FIN/ACK handshake cleanly closes the session

USE TCP WHEN...

Data integrity matters more than speed: web browsing, email, file transfers, database queries, SSH sessions. If a packet is lost, you want it retransmitted automatically.

UDP — User Datagram Protocol

UDP is a connectionless protocol — it sends packets with no handshake, no acknowledgement, and no delivery guarantee. What it lacks in reliability it makes up for in raw speed and low overhead. UDP is protocol number 17.

TCP vs UDP — Head to Head

TCP — Reliable	UDP — Fast
Connection-oriented (3-way handshake)	Connectionless (no handshake)
Ordered packet delivery	No ordering guarantee
Acknowledgements + retransmission	No retransmission mechanism
Flow & congestion control	No flow control
Higher overhead — slower	Low overhead — faster
Use case: HTTP, SSH, FTP, SMTP, RDP	Use case: DNS, DHCP, VoIP, video, NTP

USE UDP WHEN...

Speed and low latency matter more than perfection: VoIP, live video streaming, online gaming, DNS lookups, TFTP, SNMP. A dropped frame in a call is better than freezing while waiting for retransmission.

EXAM TIP

The exam often asks: 'Which protocol would you use for real-time voice communication?' The answer is always UDP. VoIP can tolerate an occasional lost packet, but cannot tolerate the delay caused by TCP retransmission. Know this distinction cold.

GRE — Generic Routing Encapsulation

GRE is a tunneling protocol (IP protocol number 47) developed by Cisco. It wraps — encapsulates — one network protocol inside another, creating a virtual point-to-point link across an IP network.

How GRE Works

GRE adds a new outer IP header and a GRE header to the original packet, allowing it to traverse networks that might not otherwise support the inner protocol:

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Original Packet:
[ IP Header  |  TCP Header  |  Data Payload ]

After GRE Encapsulation:
[ Outer IP  |  GRE Header  |  Inner IP  |  TCP  |  Data ]
   (public)     (proto 47)    (original)   ← rides inside →
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GRE Key Characteristics

- Protocol-agnostic — can tunnel IPv4, IPv6, IPX, and other protocols inside an IP network
- No built-in encryption — GRE alone is not secure; pair with IPsec for secure tunnels
- Common in VPNs — GRE + IPsec is a standard enterprise site-to-site VPN combination
- Adds overhead — extra headers increase packet size; watch for MTU fragmentation issues

GRE + IPSEC = SECURE VPN TUNNEL

GRE provides the tunnel mechanism; IPsec provides encryption and authentication. Think of GRE as a pipe with no lock — IPsec is the padlock you add for security.

Part 3 — Traffic Types

Not all network traffic is addressed the same way. The traffic type determines how many devices receive a given packet and how routers and switches handle it. There are four main types defined for the exam:

Type	Definition	Real-World Example
Unicast	One sender → one specific receiver	Most common internet traffic: web browsing, SSH, email delivery
Broadcast	One sender → all devices on subnet	ARP requests, DHCP discovery — limited to broadcast domain
Multicast	One sender → specific group of receivers	IPTV, OSPF/EIGRP routing protocols — efficient group delivery
Anycast	One sender → nearest/best receiver	CDNs, DNS root servers (e.g., Cloudflare 1.1.1.1)

Broadcast Domains vs Collision Domains

A broadcast domain is the network segment where a broadcast frame will be forwarded. Understanding the boundaries of these domains is essential for network design, VLAN configuration, and troubleshooting.

Port	Protocol	Transport	Description
Device	Breaks Collision Domain?	Breaks Broadcast Domain?	Notes
Hub	No	No	All ports share 1 collision domain and 1 broadcast domain
Switch	Yes	No	Each port = separate collision domain; broadcasts still flood all ports
Router	Yes	Yes	Routes between subnets; blocks broadcasts by default
VLAN	Yes	Yes	Creates broadcast domain boundary without a physical router

EXAM TIP

Routers break up broadcast domains. Switches break up collision domains but NOT broadcast domains. Adding a VLAN to a switch creates a new broadcast domain without a physical router. This is a very common exam scenario — know the distinction between Layer 2 and Layer 3 boundaries.

Chapter Summary & Key Takeaways

Chapter 4 covers the fundamental language that networks speak. Lock down these concepts before exam day:

- Ports are logical endpoints (0–65,535). Know the three ranges: Well-Known (0–1023), Registered (1024–49151), Ephemeral (49152–65535).
- Memorize critical port mappings: 22 (SSH), 25 (SMTP), 53 (DNS), 67/68 (DHCP), 80 (HTTP), 143 (IMAP), 443 (HTTPS), 3389 (RDP).
- ICMP is a diagnostic/error protocol — ping and traceroute use it, but it carries no user data. Protocol number 1.

- TCP is connection-oriented: three-way handshake, ordered delivery, acknowledgements, retransmission. Protocol number 6.
- UDP is connectionless: no handshake, no guarantees, low overhead, fast. Protocol number 17. Use for VoIP, video, DNS.
- GRE (protocol 47) creates tunnels by encapsulating protocols inside IP. No encryption — always pair with IPsec for security.
- Traffic types: Unicast (1→1), Broadcast (1→all in subnet), Multicast (1→group), Anycast (1→nearest). Routers break broadcast domains; switches do not.

STUDY STRATEGY

Create flashcards for every port/protocol pair in this guide. The N10-009 gives you scenario-based questions — map each protocol to a real-world failure. For example: 'User can receive but not send email' points to SMTP (port 25/587). Context is everything.

That's Chapter 4 complete. These building blocks — how data is addressed, what rules govern its journey, and how traffic is classified — are the foundation for every other networking concept ahead.

Keep labbing. Keep learning.